Maths Week 5: coordinates and straight lines (written)

Ember · Non-calculator · about 20 minutes · 11 marks

Date: _____

Skill: written method and checking. Show the numbers you substitute. When it helps, sketch a quick pair of axes and check your answer fits the picture.

[2 marks]

1. (a) Write down the coordinates of three points on the line x = −2
(b) Write down the equation of the line through (1, 4) and (6, 4).

[1 mark]

2. Where do the lines y = −3 and x = 5 cross?

Answer (_____ , _____)

AQA Maths 8300/3H, June 2018, Q8(b) **[2 marks]**

3. Is the point (−5, −6) above, below or on the line y = 3x + 7 ?
Tick one box.

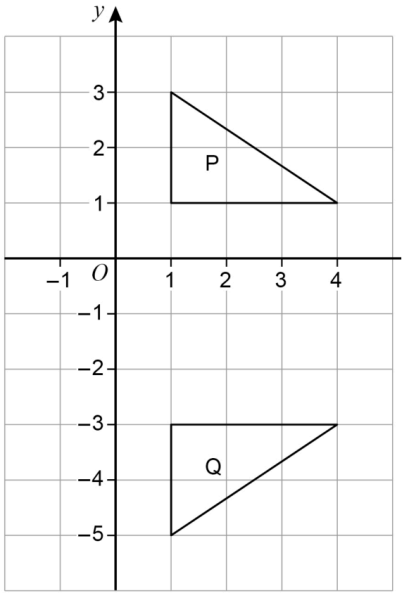
☐ **Above** ☐ **Below** ☐ **On the line**

You must show your working.
Do not use a graphical method.

Working

AQA Maths 8300/1H, November 2019, Q5(a) **[1 mark]**

4. Here are two triangles, P and Q.



AQA 8300/1H November 2019, Q5(a)

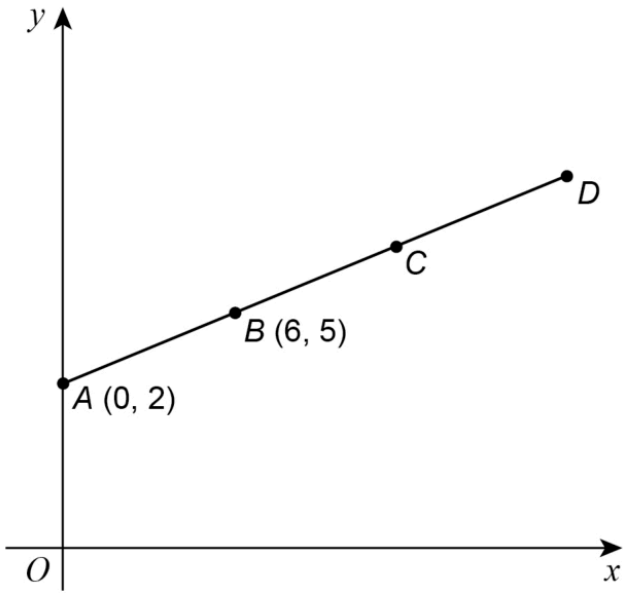
Here is a statement.

A transformation that maps P to Q is a reflection in the line x = –1

Make **one** criticism of the statement.

AQA Maths 8300/1H, June 2018, Q7 **[3 marks]**

5. *A* (0, 2) and *B* (6, 5) are points on the straight line *ABCD*.



Not drawn accurately

AQA 8300/1H June 2018, Q7

AB* = *BC* = *CD

Work out the coordinates of *D*.

Working

Answer (_____ , _____)

[2 marks]

6. Ali says, "The line x = 4 is parallel to the x-axis, because it has an x in it."
Is Ali right? Explain your answer.

Self-check (5 min)

□ Q1: my three points all have x = −2 (the first number).

□ Q3: I substituted x = −5 and wrote the line's y value before ticking.

□ Q5: I found the step from A to B, then used it twice more.

□ Q6: I said what every point on x = 4 has in common.

□ Reflection: which question did a sketch help with?

Answers and marks

- **1.** (a) Any three points with x = −2, e.g. (−2, 0), (−2, 3), (−2, −1) **(1)**. (b) y = 4 **(1)**.
- **2.** (5, −3) **(1)**. Not (−3, 5): x first.
- **3. (AQA June 2018 3H Q8(b))** When x = −5, y = 3 × (−5) + 7 = −8 **(M1)**. The point's y is −6, and −6 is bigger than −8, so **Above**, with this working **(A1)**. A tick with no working scores 0. (Worked marking: the official mark scheme for this paper is not in our archive.)
- **4. (AQA Nov 2019 1H Q5(a)), official mark scheme, B1 for any one of:** says the wrong line has been given, e.g. the line should be y = −1; or says that for the given reflection the image would be in the second quadrant (the other side of the y-axis); or says the given line is vertical (x = −1 is vertical); or gives the coordinates of an image point, e.g. (1, 1) would move to (−3, 1). **No mark** for: "it should be reflected in the y-axis", "it is not a reflection in x = −1" with no reason, or a right statement together with a wrong one.
- **5. (AQA June 2018 1H Q7), official mark scheme: M1** for a correct step, e.g. x-coordinate of D = 0 + 6 + 6 + 6 or y-coordinate of D = 2 + 3 + 3 + 3; **A1** for C = (12, 8), or D = (18, ...) or (..., 11); **A1** for (18, 11). (11, 18) does not get the last mark.
- **6.** No, Ali is wrong **(1)**: every point on x = 4 has x-coordinate 4, e.g. (4, 0) and (4, 6), so the points are straight above each other and the line is vertical, parallel to the **y**-axis **(1)**.

Score: _____ / 11

Written for Ember in AQA style. Includes AQA Mathematics 8300/3H June 2018 Q8(b), 8300/1H November 2019 Q5(a), 8300/1H June 2018 Q7; on screen 8300/2H November 2022 Q3, 8300/3H June 2018 Q19, 8300/2H November 2018 Q2. © AQA, used for private practice.